

SWARUP G L

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Summary

Artificial Intelligence and Data Science undergraduate specializing in Medical Image Processing, Deep Learning Architectures, and Multi-modal Sensor Fusion. Experience developing Edge AI solutions, Internet of Things (IoT) hardware, and Network-Centric 6G Integrated Sensing and Communication (ISAC) frameworks. Seeking to leverage expertise in deterministic Ultra-Reliable Low-Latency Communication (URLLC), raw byte stream analysis, and computer vision for advanced scientific research.

Education

Indian Institute of Information Technology (IIIT)

2023 – 2027

B.Tech in Artificial Intelligence and Data Science

Raichur, Karnataka

Technical Skills

AI & Computer Vision: PyTorch, TensorFlow, CNNs, LSTMs, Ordinal Regression, MobileNet, VGG, Inception

Languages: Python, JavaScript, C, C++, MATLAB

IoT Tools & Hardware: Arduino IDE, ESP8266, Raspberry Pi, DWM1000 UWB, Neo-6M GPS, Biometric Sensors

Backend & Web Technologies: Node.js, Express.js, MongoDB, IPFS, Blockchain

Current Research & Experience

Ghost-Eye: Network-Centric 6G ISAC Framework

Jan 2026 – Present

Principal Researcher / Core System Architect

Academic Project

- Designing a 6G Integrated Sensing and Communication (ISAC) framework for Non-Line-of-Sight (NLOS) perception and deterministic sub-millisecond accident response utilizing joint bistatic Terahertz OFDM waveforms (0.1–1 THz).
- Modeled a Two-Layer Hybrid Resource Allocation architecture using Lyapunov drift-plus-penalty optimization to isolate a URLLC network slice, dynamically prioritizing emergency post-crash traffic to achieve a 67% P99 latency reduction.

Bosch Global Software Technologies (BGSW) University Engagement

Real Time Location System (RTLS) Using DWM-1000 Ultra-wideband Module

Corporate Collaboration

- Developed a high-precision indoor positioning system using UWB technology, implementing a functional array with a tight sub-15 cm error rate for low-power, localized asset deployment tracking.
- Engineered physical telemetry tags combining DWM1000 modules with ESP8266 microcontrollers, scheduling continuous communication updates via a customized C++ Arduino IDE environment over Wi-Fi channels.
- Parsed and filtered live tracking arrays using the MQTT protocol (HiveMQ client) to stream spatial coordinates to cloud infrastructure, handling multipath non-line-of-sight signal reflections and structural scattering profiles.

Academic Projects

Bluetooth-Controlled Biometric Student Attendance | *ESP8266, Fingerprint Module, Mobile App*

- Developed a wireless biometric attendance architecture interfacing a fingerprint scanner with an ESP8266.
- Integrated the hardware with a custom mobile application via Bluetooth to manage, authorize, and securely log real-time student attendance data directly from the device.

MalConv USB Scanner: Gated CNN for Malware Detection | *PyTorch, Python, CNNs*

- Architected an end-to-end MalConv-style gated Convolutional Neural Network to score Windows executables and packed/obfuscated binaries directly from raw byte streams, bypassing traditional malware signatures.
- Engineered a learned attention-like gating mechanism utilizing parallel convolutions and sigmoid activations to suppress noise and selectively pass salient structural features.
- Developed a functional deployment pipeline including automated USB monitoring, recursive zip-archive sanitization, and real-time malicious file quarantine protocols.

Verdict Vault – Secure Distributed Document Storage | *Blockchain, IPFS, MongoDB, Node.js*

- Architected a secure distributed record network utilizing immutable database ledgers and IPFS secure file hashing to securely connect and synchronize records between law enforcement, advocates, and judicial officers.

Dormant Device Detection and Tracking System | *ESP8266, GPS, MOSFETs, App Integration*

- Designed a telemetry platform to track system components, implementing an automated 10% power hardware switchover loop to optimize operational lifespan while streaming geographical coordinate data packs.

GPS-Based Turn Alert & Speed Suggestion System | *Neo-6M GPS, ESP8266*

- Engineered a real-time tracking system that parses source-to-destination GPS coordinates to identify upcoming topological curves and automatically calculate safe speed recommendations based on geometric curve angles.
- Integrated automated proximity warnings triggered at precise geographic distances before approaching the turn.

Real-Time Performance Tracking Platform (Leaderboard) | *JavaScript, Node.js, Express.js*

Ongoing

- Building a scalable web backend utilizing clean REST API routes and database indexing schemas to process, sort, and display rapid running telemetry data metrics on a live interface.

Medical Imaging & Deep Learning Specialized Coursework

Chrono-Aware Embryo Phase Classification | *PyTorch, CNN + LSTM, Hybrid Loss*

- Developed a 16-class time-lapse sequence model using a MobileNetV2 feature extractor and LSTM backend to track sequential biological growth across critical morphological boundaries.
- Engineered a custom Hybrid Ordinal Regression Loss blending cross-entropy with a CDF SmoothL1 ordinal penalty to enforce chronological consistency and mitigate overfitting gaps down to 0.15 percentage points.
- Achieved an 89.42% tolerance accuracy (± 1 phase) and 67.00% exact target classification on sequential video datasets.

Breast Ultrasound Tumor Classification | *PyTorch, Transfer Learning*

- Architected a highly regularized medical pipeline to classify Breast Ultrasound Images into Normal, Benign, and Malignant.
- Designed a custom 48K-parameter Depthwise Separable CNN trained from scratch to isolate spatial filtering from channel mixing, compressing the mathematical model parameter footprint by 99.96% relative to VGG16 baselines.
- Achieved an 83.33% peak testing accuracy while applying Batch Normalization and Global Average Pooling to entirely mitigate structural texture memorization.

Thermal Foot Ulcer Detection: DIP vs. CNN | *OpenCV, Computer Vision, CNN*

- Built a comparative diagnostic system pairing traditional digital image processing with convolutional neural networks for automated non-contact Diabetic Mellitus screening from thermographic footage.
- Developed localized ROI segmentation pipelines in the HSV color space, using Hue boundaries and morphological smoothing to isolate critical red hotspots and construct a 5-channel feature map, achieving a final classification precision of 80.4%.

Hackathons & Competitions

AUTown'24 Project Competition | *BITS Pilani, Hyderabad Campus*

Nov 2024

Core Engineering Department (MADE Theme)

- Participated in a national project competition focused on Advanced Driver Assistance Systems (ADAS) technology hosted across the BITS Pilani Work Integrated Learning Programmes network.

Extracurricular Experience

ElectroGeeks Club

Jan 2025 – Feb 2026

Coordinator

IIT Raichur

- Designed and supervised technical club workshops focusing on embedded microcontrollers, drone building architectures, real-time telemetry calibration, and multi-component hardware debugging sessions for junior peers.
- Managed campus-wide hardware deployment operations, technical component inventory tracking, and electronic procurement setups.